

Homework 1 (CTL): ComS/CprE/SE 412, ComS 512

Due-date: Feb 9 at 11:59PM.

Late submission policy. If you submit within one day after the deadline then there will be 10% penalty. That is, if your score is x points, then your final score for this homework after the penalty will be $0.9 \times x$.

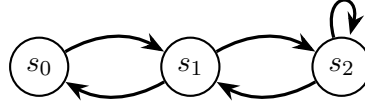
Submissions that are beyond one day late will not be graded without (prior) explicit permission from the instructors.

Submission format. Your submission file must be pdf format. You will use canvas-gradescope for submission (this will allow you to associate scanned pages/pdfs of your submission with the question number). Here is a short video on how to submit homework using gradescope

<https://www.youtube.com/watch?v=u-pK4GzpId0>

You are encouraged to use latex to prepare your submission but this is not a requirement.

1. Consider the following Kripke structure, with $p \in L(s_0) \cap L(s_2)$ and $q \in L(s_2)$.



Identify the set of states that satisfy each of the following:

- (a) $EX(q)$
- (b) $AX(p)$
- (c) $AX(q)$
- (d) $AG(p)$
- (e) $EG(p)$
- (f) $AF(p)$
- (g) $AG(EX(p))$
- (h) $AG(AF(p))$

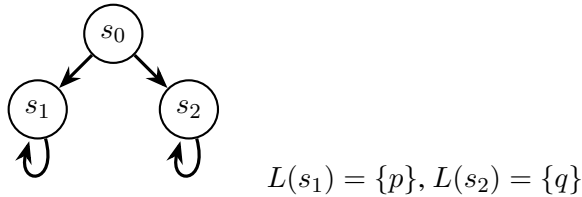
(16 pts)

2. Express the following statements as CTL formula:

(4+4+6 pts)

- (a) Along all paths **withdraw-money** is never true after **invalid-login**.
- (b) Along all execution sequences of an elevator behavior, if the elevator door is **open** then the door remains **open** until a **request-to-move** is sent to the elevator.
- (c) Whenever proposition **request-site-update** is true in a state, it is followed in zero or more steps by a state where proposition **updating-site** is true, which in turn is followed in one or more steps by a state where **update-complete** is true.

3. To disprove that two CTL formula are equivalent, you are required to draw a Kripke structure and identify a state in that structure, which satisfies one of the formula and does not satisfy the other. For instance, in order to disprove that $EX(p) \wedge EX(q)$ and $EX(p \wedge q)$ are equivalent, we can draw the following the Kripke structure:



We present the labels for the relevant states in the Kripke structure and state that s_0 satisfies $EX(p) \wedge EX(q)$ and s_0 does not satisfy $EX(p \wedge q)$. This is because there exists a path from s_0 where in the very next state s_1 the proposition p is true; thus conforming that s_0 satisfies $EX(p)$ (similarly, one can justify the satisfiability of $EX(q)$ at state s_0). On the other hand, there exists no path from s_0 , where in the very next state both p and q are satisfied.

Disprove that the two CTL formula $AF(p \wedge q)$ and $AF(p) \wedge AF(q)$ are equivalent.

(5 pts)

4. To prove that one formula (say, φ_1) is “stronger” than another (say, φ_2), you need to prove that whenever in any state in any Kripke structure φ_1 holds, φ_2 holds in that state as well. In other words, φ_1 is “stronger” than φ_2 if and only if $\varphi_1 \Rightarrow \varphi_2$ is a tautology (always evaluates to true). For instance, to prove that $AX(p)$ is stronger than $EX(p)$ we can write:

$$\begin{aligned} \forall s. s \in [AX(p)] &\Leftrightarrow \forall \pi \in Path(s) : \pi[1] \in [p] \\ &\Rightarrow \exists \pi \in Path(s) : \pi[1] \in [p] \\ &\Rightarrow s \in [EX(p)] \end{aligned}$$

To disprove that $EX(p)$ is stronger than $AX(p)$, you will draw a Kripke structure and present a state in the Kripke structure that satisfies $EX(p)$ but does not satisfy $AX(p)$ (see the Kripke structure example in the previous problem).

Prove or disprove $AG(p \wedge q)$ is stronger than $AG(p \Rightarrow AF(q))$.

(5 pts)

5. **Extra Credit for 412; Required problem for 512.** We will define a new operator $A\circ$ as follows. A state s satisfies $A(\varphi_1 \circ \varphi_2)$ if and only if for all paths starting from s at least one of the following holds

- there exists a state where φ_2 is satisfied and before that φ_1 holds in all states in the path
- all states in the path satisfy φ_1 and not φ_2

Prove or disprove that

- (a) $A(p \circ \text{false})$ can be expressed in CTL.
- (b) $A(\text{false} \circ p)$ can be expressed in CTL.

(10pts)